BLUETOOTH CAR-KIT BOARD - DRAFT SCHEMATIC (generated, unreviewed). Notes: electronics/bluetooth/README.md.
One USB cable from the H5 -> USB2514Bi hub -> MCP2221A (AT commands to the Feasycom FSC-BT1026C), PCM2900C USB codec (records the module line out: phone music and calls go into the H5), and the module own USB.

USB in, protection, 3.3 V

File: camry-bt-usb.kicad_sch

USB hub

File: camry-bt-hub.kicad_sch

USB-serial for AT commands

File: camry-bt-serial.kicad_sch

Bluetooth module, power-on, LEDs, mic

File: camry-bt-module.kicad_sch

Audio into the H5 (USB audio codec)

File: camry-bt-audio.kicad_sch

DRAFT generated by electronics/bluetooth/kicad/gen_bt.py - not reviewed, not built

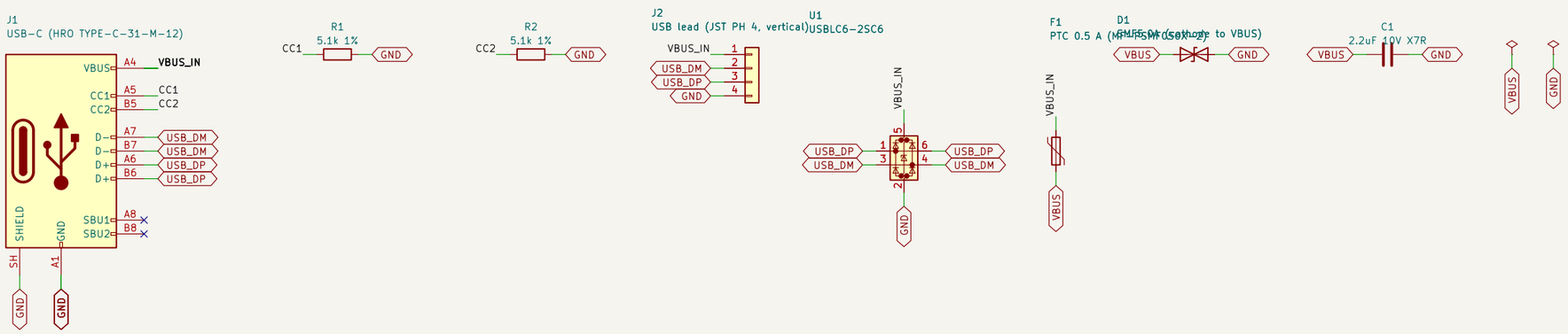
What If Garage - Drag Monster Camry

Sheet: /
File: camry-bt.kicad_sch

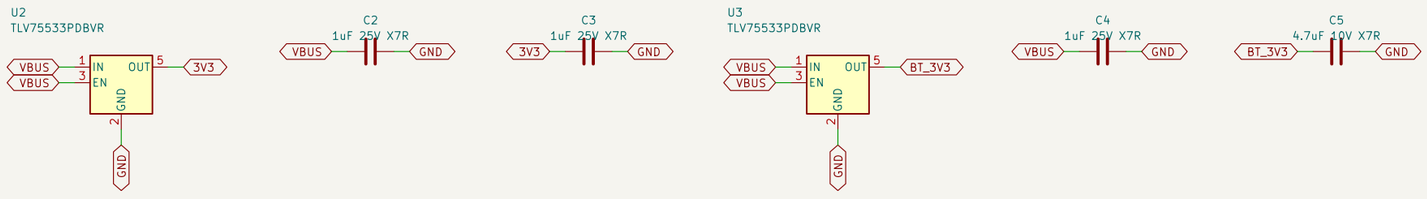
Title: Bluetooth car-kit board

Size: A3	Date: 2026-11-09	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 1/6

UPSTREAM USB 2.0 FROM THE H5: USB-C socket OR a JST PH 4 lead (crimped to the H5 header port P6); fit one cable only



3.3 V REGULATORS (TI TLV755, 500 mA): one for the hub and the USB-serial chip, one for the Bluetooth module (kept apart from the hub noise)



Every pin is connected by net labels; KiCad ERC + check.py verify connectivity
DRAFT generated by electronics/bluetooth/kicad/gen_bt.py — not reviewed, not built

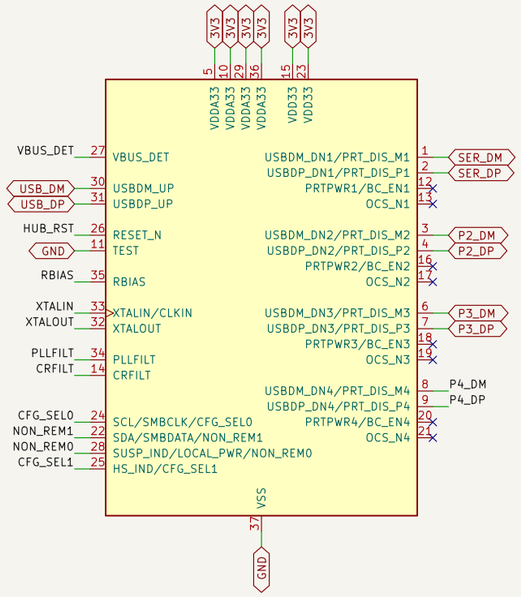
What If Garage — Drag Monster Camry

Sheet: /USB in, protection, 3.3 V/
File: camry-bt-usb.kicad_sch

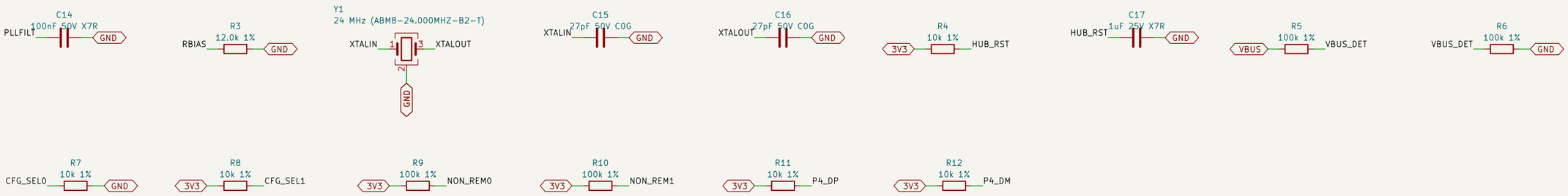
Title: Bluetooth car-kit board — USB in, protection, 3.3 V

Size: A3	Date: 2026-11-09	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 2/6

USB 2.0 HUB Microchip USB2514Bi: port 1 = USB-serial (AT commands), port 2 = USB audio codec, port 3 = the module's own USB, port 4 off







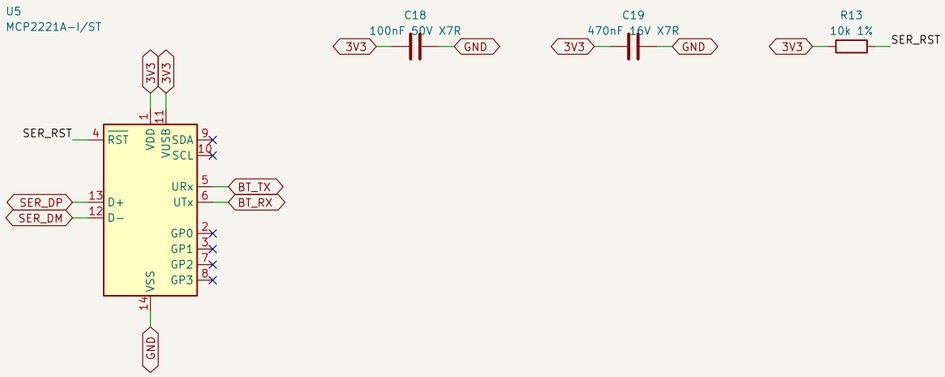
Every pin is connected by net labels; KiCad ERC + check.py verify connectivity
DRAFT generated by electronics/bluetooth/kicad/gen_bt.py — not reviewed, not built
What If Garage — Drag Monster Camry

Sheet: /USB hub/
File: camry-bt-hub.kicad_sch

Title: Bluetooth car-kit board — USB hub

Size: A3	Date: 2026-11-09	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 3/6

USB–SERIAL Microchip MCP2221A (USB CDC, no driver on Windows 10/11): the module's AT–command UART, 115200 8N1, 3.3 V levels



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DRAFT generated by electronics/bluetooth/kicad/gen_bt.py — not reviewed, not built

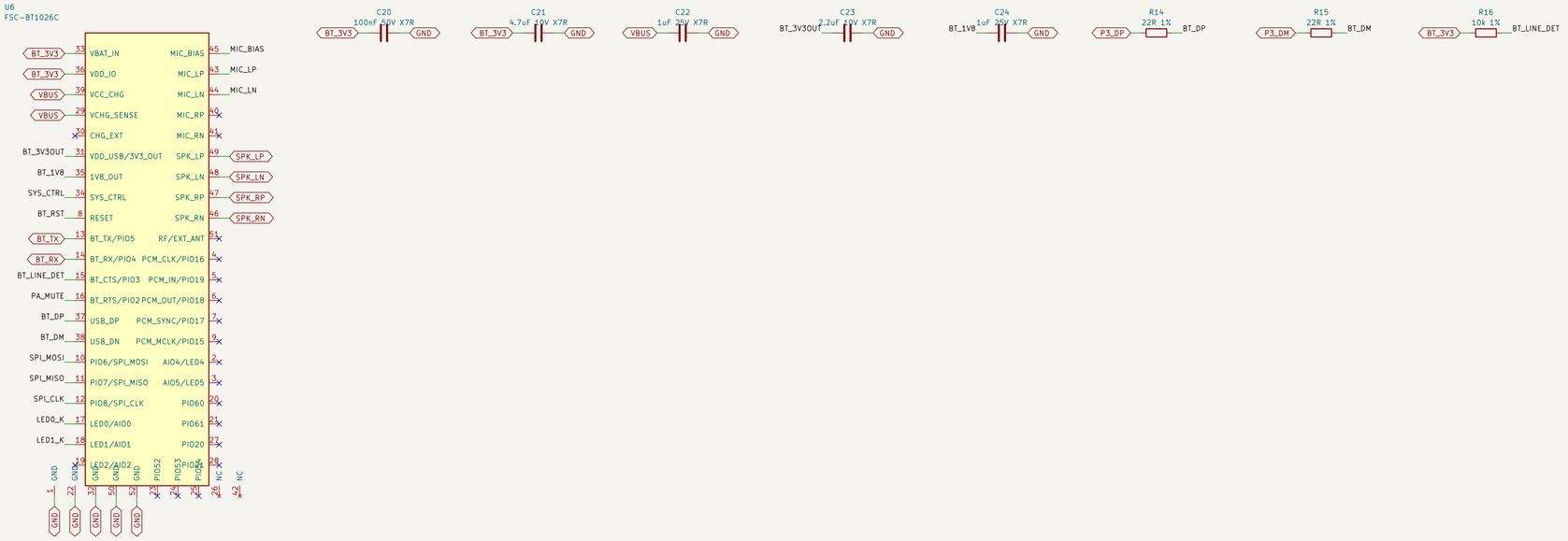
What If Garage — Drag Monster Camry

Sheet: /USB–serial for AT commands/
File: camry–bt–serial.kicad_sch

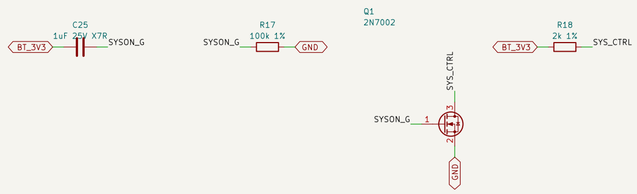
Title: Bluetooth car–kit board — USB–serial for AT commands

Size: A3	Date: 2026–11–09	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 4/6

BLUETOOTH MODULE Feasycom FSC-BT1026C: powered from BT_3V3 (VBAT_IN, VDD_IO); VCC_CHG on 5 V for its USB (charger stays off, its default)



POWER-ON: Feasycom "Option One" delayed start (SYS_CTRL low > 100 ms after BT_3V3, then high), 2N7002 for their 2SK3018



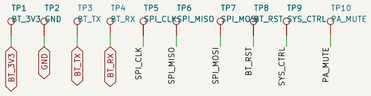
STATUS LEDS on the module's open-drain LED pads (tolerant to 7 V, so fed from 5 V): LED0 blinks while pairing, LED1 on when connected



EXTERNAL CABIN MICROPHONE on MIC_L (the input calls use): electret, biased from MIC_BIAS (Feasycom reference "MIC")



TEST PADS: Feasycom update / debug (VDD, SPI, UART) and the spare control pins



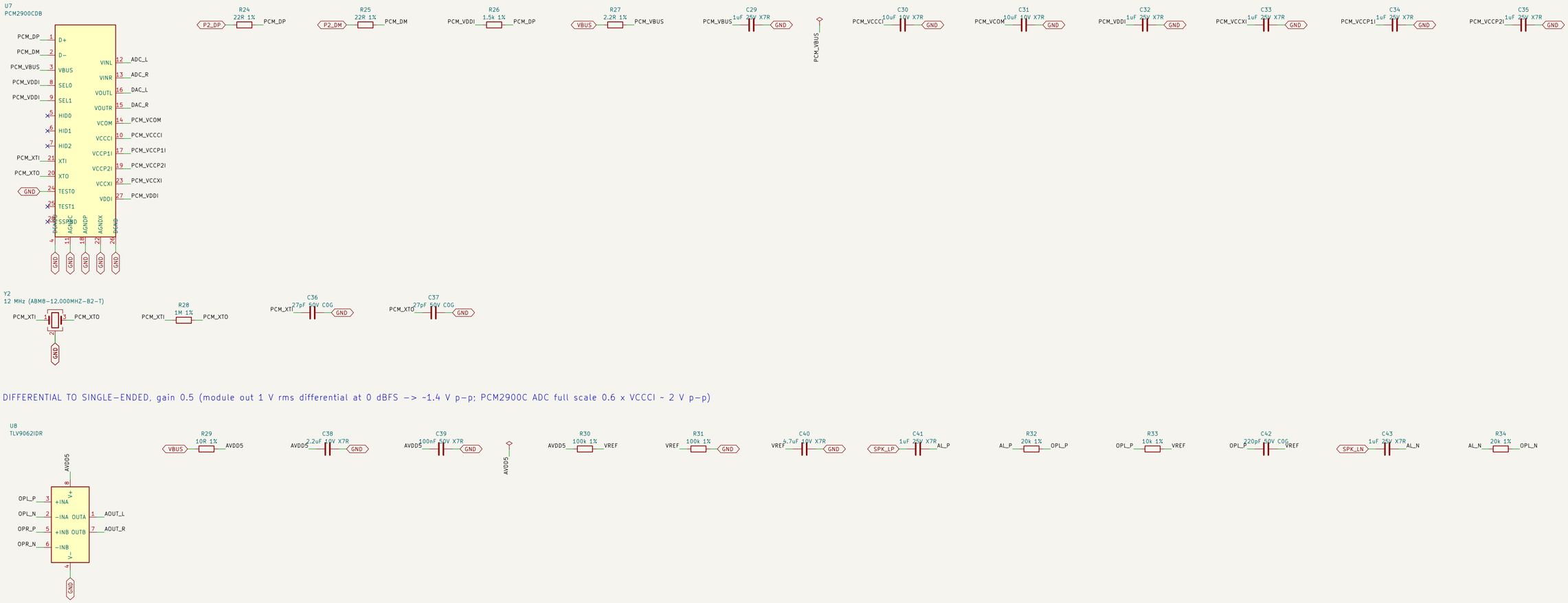
Every pin is connected by net labels; KiCad ERC + check.py verify connectivity
DRAFT generated by electronics/bluetooth/kicad/gen_bt.py — not reviewed, not built
What If Garage — Drag Monster Camry

Sheet: /Bluetooth module, power-on, LEDs, mic/
File: camry-bt-module.kicad_sch

Title: Bluetooth car-kit board — Bluetooth module, power-on, LEDs, mic

Size: A2	Date: 2026-11-09	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 5/6

USB AUDIO CODEC TI PCM2900C, bus powered (SBFS039 Figure 38): the H5 records the phone audio from its ADC



DIFFERENTIAL TO SINGLE-ENDED, gain 0.5 (module out 1 V rms differential at 0 dBFS -> ~1.4 V p-p; PCM2900C ADC full scale 0.6 x VCCCI ~ 2 V p-p)



SPARE LINE OUT from the codec's DAC (the H5 can play into it: bench headphones, or prompts into a call later)



Every pin is connected by net labels: KiCad ERC + check.py verify connectivity
DRAFT generated by electronics/bluetooth/kicad/gen_bt.py — not reviewed, not built

What If Garage — Drag Monster Camry

Sheet: /Audio into the H5 (USB audio codec)/
File: camry-bt-audio.kicad_sch

Title: Bluetooth car-kit board — Audio into the H5 (USB audio codec)

Size: A2	Date: 2026-11-09	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 6/6